CT-2 Memorize List

CT-2, Thu 5 Nov 2026, 9–10 am, NR321. The whole syllabus in 35 cards. Each card: **Formula** → **Use when** (how the question looks) → **How to plug** → **Caveats** (when it does NOT work) → **Example** (a past-paper question, worked step by step).

1. Filtrations and martingales

1 Filtration, adapted, measurable: what are they and how do I answer "is Y $\mathcal{F}_n$-measurable?"

FORMULA

Filtration: $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$. Natural filtration: $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$. Adapted: X_t is $\mathcal{F}_t$-measurable for every t.

USE WHEN

"Is Y $\mathcal{F}_n$-measurable?", "Is the process adapted?", or as condition (i) of a martingale.

HOW TO PLUG

1. Write Y using the X_k's.
2. Only $X_1, \ldots, X_n$ appear → Yes. Any X_{n+1} or later appears → No.

CAVEATS

- Any function of adapted things is adapted (S_n^2, e^{S_n}, $\max_{k \leq n} S_k$).
- A future increment (ξ_{n+1}, $W_{t+h} - W_t$) is never $\mathcal{F}_n$- / $\mathcal{F}_t$-measurable.

EXAMPLE (PRACTICE)

$S_n = \xi_1 + \cdots + \xi_n$. Is S_{n+1} $\mathcal{F}_n$-measurable?
Step 1: $S_{n+1} = S_n + \xi_{n+1}$. Step 2: ξ_{n+1} is a future step → **No**. Is S_n^2? It uses only $\xi_1 .. \xi_n$ → **Yes**.

2 Martingale: the definition, sub and super, and what follows from it.

FORMULA

(i) adapted, (ii) $E|X_t| < \infty$, (iii) $E[X_{t+1} \mid \mathcal{F}_t] = X_t$.
Sub-MG: $E[X_{t+1} \mid \mathcal{F}_t] \geq X_t$ (goes up). Super-MG: $\leq$ (goes down).
Consequences: $E[X_{t+s} \mid \mathcal{F}_t] = X_t$ and $E[X_t] = E[X_0]$ for all t.

USE WHEN

"Is X_n a martingale? (Yes/No)", "Which statement is TRUE?", "sub or super?".

HOW TO PLUG

1. State (i) and (ii) in one line each.
2. Compute $E[X_{t+1} \mid \mathcal{F}_t]$ (card 3) and compare with X_t: equal → MG, bigger → sub, smaller → super.

CAVEATS

- Constant mean is necessary, NOT sufficient: $EX_t = EX_0$ does not prove MG.
- "Super" goes DOWN (the casino's game). Names are counter-intuitive.

EXAMPLE — CT-1'26 Q8 (ALL SETS)

"Which statement is TRUE?" → pick the option "the mean of a martingale process is constant with respect to time", because $E[X_t] = E[X_0]$.

3 How to compute $E[X_{n+1} \mid \mathcal{F}_n]$ in one line.

FORMULA

If A is known at time n and B is independent of $\mathcal{F}_n$: $E[A \cdot B \mid \mathcal{F}_n] = A \cdot E[B]$, $E[A + B \mid \mathcal{F}_n] = A + E[B]$.

USE WHEN

Every discrete "is it a martingale?" question.

HOW TO PLUG

1. Write X_{n+1} = (something known at n) combined with (the new step).
2. Pull the known part out.
3. Replace the new step by its mean.
4. Compare with X_n.

CAVEATS

Step 3 needs the new step to be independent of the past (iid steps: fine).

EXAMPLE — CT-2'25 Q3

$X_n = 2^n U_1 \cdots U_n$, $U_i \sim U(0,1)$ iid. MG?

Step 1: $X_{n+1} = X_n \cdot 2U_{n+1}$. Step 2: $E[X_{n+1} \mid \mathcal{F}_n] = X_n \, E[2U_{n+1}]$. Step 3: $E[2U] = 2 \cdot \frac{1}{2} = 1$. Step 4: $= X_n$ → **Yes, MG**.

4 Random-walk martingales: $S_n = Y_1 + \cdots + Y_n$, Y_i iid with mean μ, variance σ^2.

FORMULA

Process	Martingale when
S_n	$\mu = 0$
$S_n - n\mu$	always
$(S_n - n\mu)^2 - n\sigma^2$	always (so $S_n^2 - n\sigma^2$ when $\mu = 0$)
$X_0 Y_1 Y_2 \cdots Y_n$	$E[Y] = 1$
$e^{\theta S_n} / \big(E[e^{\theta Y}]\big)^n$	always

USE WHEN

"X_n is a martingale if $E(Y_1) =$ __", walk or product of iid steps.

HOW TO PLUG

1. Decide: sum or product.
2. Find $\mu = E[Y]$ (and σ^2 if squared).
3. Read the row's condition.

CAVEATS

- $S_n^2 - n\sigma^2$ needs $\mu = 0$; otherwise centre first.
- If $\mu > 0$, S_n is a sub-MG; if $\mu < 0$, super-MG.

EXAMPLE — MIDSEM'24 Q1(B)
"$X_n = \sum_{i \le n} Y_i$ is a martingale if $E(Y_1) =$ __" → sum, row 1 → 0.

2. Brownian motion and related processes

5 Wiener process: axioms, and the 4-point test "is $X(t)$ a Wiener process?".

FORMULA

Axioms: $W(0) = 0$; independent increments; $W(t) - W(s) \sim N(0, t-s)$; continuous paths.
Test: (1) $X(0) = 0$, (2) Gaussian, (3) $EX(t) = 0$, (4) $\mathrm{Cov}(X_s, X_t) = \min(s,t)$.

USE WHEN

"Short note on the Wiener process"; T/F "$V(t) = \ldots$ is a Wiener process".

HOW TO PLUG

1. Check $X(0) = 0$ first (quickest fail).
2. Gaussian? (card 6: $f(t)W(g(t))$ always is.)
3. Compute the covariance and compare with $\min(s,t)$.

CAVEATS

Covariance $e^{-\lambda|t-s|}$ (depends on $|t-s|$ only) means stationary OU, NOT Wiener.

EXAMPLE — MIDSEM'25 Q2
"Short note on the Wiener process" → write the 4 axioms, then add $\mathrm{Cov}(W_s, W_t) = \min(s,t)$ and "paths continuous but nowhere differentiable".

6 ⭐ Master formula: $X(t) = f(t)\,W(g(t))$ with f, g deterministic. Gaussian? Covariance? Wiener?

FORMULA

Always Gaussian, mean 0, $\quad \mathrm{Cov}(X_s, X_t) = f(s)f(t)\min\big(g(s), g(t)\big)$.
Wiener $\iff$ that equals $\min(s,t)$ and $X(0) = 0$.

USE WHEN

"Is $X(t) = e^{-t}W(e^{2t})$ Gaussian?", "Is $\frac{1}{c}W(c^2t)$ / $tW(1/t)$ a Wiener process?", "Cov of X?".

HOW TO PLUG

1. Read off f and g.
2. Gaussian question → Yes immediately.
3. Wiener question → compute $f(s)f(t)\min(g(s), g(t))$ and $X(0) = f(0)W(g(0))$.

$X(t)$	f, g	Cov	Wiener?
$\frac{1}{c}W(c^2t)$	$\frac{1}{c}$, c^2t	$\min(s,t)$	Yes
$tW(1/t)$	t, $\frac{1}{t}$	$\frac{st}{\max(s,t)} = \min(s,t)$	Yes
$e^{-\lambda t}W(e^{2\lambda t})$	$e^{-\lambda t}$, $e^{2\lambda t}$	$e^{-\lambda\|t-s\|}$	No (OU)

CAVEATS

- f and g must be non-random (no W inside them), and $g \geq 0$.
- Gaussian ≠ Wiener: $e^{-t}W(e^{2t})$ has $X(0) = W(1) \neq 0$.

EXAMPLE — CT-1'26 Q3

"$X(t) = e^{-t}W(e^{2t})$ is a Gaussian process (Yes/No)" → Step 1: $f = e^{-t}$, $g = e^{2t}$, both deterministic. Step 2: → **Yes**, no computation.

7 The 4 transforms of BM that are again standard BM.

FORMULA

$\frac{1}{c}W(c^2t)$ (scaling), $-W(t)$ (symmetry), $tW(1/t)$ with 0 at $t = 0$ (time inversion), $W(T) - W(T - t)$ on $[0, T]$ (time reversal).

USE WHEN

T/F "… is also a Wiener process".

HOW TO PLUG

1. Match the expression to a row → True.
2. No match → run card 6 / card 5 test.

CAVEATS

$cW(t)$ with $c \neq \pm 1$ is NOT standard BM (variance c^2t). Scaling needs the $\frac{1}{c}$ outside AND c^2 inside.

EXAMPLE — MIDSEM'25 Q1(E)

"$V(t) = \frac{1}{c}W(c^2t)$, $c > 0$, is a Wiener process (T/F)" → scaling row → **True**. Check: $V(0) = 0$, $\text{Cov} = \frac{1}{c^2}\min(c^2s, c^2t) = \min(s, t)$ ✓.

8 Lévy's characterization and the path facts (short notes).

FORMULA

Lévy: X continuous, $X_0 = 0$, X_t and $X_t^2 - t$ both MGs $\Rightarrow$ standard BM.
Paths: continuous, nowhere differentiable; quadratic variation $\sum(\Delta W)^2 \to t$, i.e. $(dW)^2 = dt$; total variation $= \infty$; Markov.

USE WHEN

"Short note", "explain quadratic variation", "prove the limit is BM".

HOW TO PLUG

1. Write the facts as sentences.
2. Add the reason: $\Delta W \sim \sqrt{\Delta t}$, so $(\Delta W)^2 \sim \Delta t$ survives.

CAVEATS

Quadratic variation is finite ($= t$) while total variation is infinite: don't mix them up.

EXAMPLE — ENDSEM-STYLE

"Explain quadratic variation of BM" → $\sum_j (W_{t_j} - W_{t_{j-1}})^2 \to t$ as the mesh $\to 0$, so $(dW)^2 = dt$; this is why Itô's formula has the extra $\frac{1}{2} f'' \, dt$ term.

9 ☆ Distribution of any linear combination $\sum a_i W_{t_i}$.

FORMULA

$$\sum_i a_i W_{t_i} \sim N\Big(0,\ \sum_i a_i^2 t_i + 2\sum_{i<j} a_i a_j \min(t_i, t_j)\Big)$$

Two terms: $\mathrm{Var}(aW_s + bW_t) = a^2 s + b^2 t + 2ab\min(s,t)$. $\mathrm{Corr}(W_s, W_t) = \sqrt{s/t}$, $s \le t$.

USE WHEN

"$4W_1 + 6W_3 \sim$?", "$\mathrm{Var}(\ldots)$", "$\mathrm{Cov}(W_1, W_4)$".

HOW TO PLUG

1. List coefficients a_i and times t_i.
2. Square terms: $a_i^2 t_i$.
3. Cross terms: $2a_i a_j \times$(smaller time).
4. Add → variance; mean is 0.

CAVEATS

Mean 0 only for standard BM from 0. With drift μt or start x_0, add those to the mean (variance unchanged).

EXAMPLE — CT-1'26 Q4

$4W_1 + 6W_3 \sim$? → Step 1: $a = (4, 6)$, $t = (1, 3)$. Step 2: $16(1) + 36(3) = 124$. Step 3: $2(4)(6)\min(1,3) = 48$. Step 4: $N(0, 172)$.

Midsem'25 Q1(d): $W_1 - W_2 + 3W_4$ → squares $1 + 2 + 36 = 39$; cross $2[(-1)(1) + (3)(1) + (-3)(2)] = -8$ → $N(0, 31)$.

10 ☆ Any mixed moment $E[g(W_s)h(W_t)]$, $s < t$: increment splitting.

FORMULA

$W_t = W_s + \Delta$, $\Delta \sim N(0, t-s)$ independent of W_s.

Even powers of a normal (double factorial): if $X \sim N(0, v)$,

$$E[X^{2k}] = (2k-1)!!\, v^k, \qquad (2k-1)!! = 1 \cdot 3 \cdot 5 \cdots (2k-1) = \frac{(2k)!}{2^k\, k!}, \qquad E[X^{2k+1}] = 0$$

k	$E[Z^{2k}]$, $Z \sim N(0,1)$	$E[W_t^{2k}]$	$E[\Delta^{2k}]$, $\Delta \sim N(0, t-s)$
1	$1!! = 1$	t	$t - s$
2	$3!! = 1 \cdot 3 = 3$	$3t^2$	$3(t-s)^2$
3	$5!! = 1 \cdot 3 \cdot 5 = 15$	$15t^3$	$15(t-s)^3$
4	$7!! = 1 \cdot 3 \cdot 5 \cdot 7 = 105$	$105t^4$	$105(t-s)^4$

Results: $E[W_s W_t] = \min(s,t)$, $E[W_s^2 W_t^2] = s(2s + t)$ for $s \le t$.

USE WHEN

"Compute $E(W_2^2 W_3^2)$", "$E[W_2 W_3^3]$", any product of BM at two times.

HOW TO PLUG

1. Let s = the smaller time. Write $W_t = W_s + \Delta$.
2. Expand the product.
3. Delete terms with an odd power of Δ or of W_s.
4. Use independence: $E[W_s^k \Delta^m] = E[W_s^k] E[\Delta^m]$.

CAVEATS

s must be the SMALLER time; Δ is "later minus earlier".

EXAMPLE — CT-1'26 Q2

$E(W_2^2 W_3^2) \to s = 2$, $\Delta = W_3 - W_2 \sim N(0,1)$.

$E[W_2^2 (W_2 + \Delta)^2] = EW_2^4 + 2E[W_2^3] E\Delta + EW_2^2 \, E\Delta^2 = 3(4) + 0 + 2(1) = 14$. Shortcut: $s(2s+t) = 2(7) = 14$ ✓.

11 ☆ Exponential of a normal: $E[e^X]$ and $\mathrm{Cov}(e^{W_s}, e^{W_t})$.

FORMULA

$$X \sim N(m, v) \Rightarrow E[e^X] = e^{m + v/2}, \ \mathrm{Var}(e^X) = e^{2m+v}(e^v - 1).$$

$$\mathrm{Cov}(e^{W_s}, e^{W_t}) = e^{(s+t)/2}\left(e^{\min(s,t)} - 1\right)$$

USE WHEN

Any $E[e^{\cdots}]$ with BM inside; "$Y_t = e^{W_t}$, find $\mathrm{Cov}(Y_1, Y_2)$".

HOW TO PLUG

1. Write the exponent as a normal: find its mean m and variance v (card 9).
2. $E[e^X] = e^{m + v/2}$.
3. For Cov: $E[e^{W_s + W_t}] - E[e^{W_s}] E[e^{W_t}]$.

CAVEATS

Exponent must be normal (a linear combination of W's). $E[e^{W_t^2}]$ is NOT this formula.

EXAMPLE — CT-1'26 Q9

$Y_t = e^{W_t}$, $\mathrm{Cov}(Y_1, Y_2) \to s = 1$, $t = 2$: $e^{(1+2)/2}(e^1 - 1) = e^{3/2}(e - 1)$.

Rebuild: $W_1 + W_2 \sim N(0, 1 + 2 + 2) = N(0, 5)$, so $E[e^{W_1 + W_2}] = e^{5/2}$; minus $e^{1/2} e^1 = e^{3/2}$; $e^{5/2} - e^{3/2} = e^{3/2}(e - 1)$ ✓.

12 ☆ Is $f(t, W_t)$ a martingale? The one test.

FORMULA

$$f(t, W_t) \text{ is a MG} \iff \partial_t f + \tfrac{1}{2} \partial_{ww} f = 0$$

Family $e^{\lambda t} g(W_t)$: MG $\iff \lambda g + \tfrac{1}{2} g'' = 0$. For $\sin$, $\cos$: $\lambda = \tfrac{1}{2}$ exactly.

$f(t, W_t)$	$\partial_t f + \frac{1}{2}\partial_{ww} f$	MG?
W_t, $W_t^2 - t$, $W_t^3 - 3tW_t$	0	Yes
$e^{\lambda W_t - \lambda^2 t/2}$	0	Yes
$e^{t/2}\sin W_t$, $e^{t/2}\cos W_t$	0	Yes
e^{W_t}, $e^t W_t$, $e^t \cos W_t$	$\neq 0$	No

USE WHEN

"Is $Y_t = \ldots$ a martingale? (Yes/No)" with Y a function of t and W_t.

HOW TO PLUG

1. Compute $\partial_t f$ and $\partial_{ww} f$.
2. Form $\partial_t f + \frac{1}{2}\partial_{ww} f$.
3. Zero → Yes. Not zero → No.

CAVEATS

- Only for functions of (t, W_t). For a general Itô process use card 25 (drift of dX).
- Watch negated statements: "$|W_t|^2 - t$ is NOT a MG" is **False**.

EXAMPLE — CT-1'26 Q5

"Is $Y_t = e^{t/2}\sin W_t$ a MG?" → Step 1: $\partial_t f = \frac{1}{2}f$, $\partial_{ww} f = -f$. Step 2: $\frac{1}{2}f - \frac{1}{2}f = 0$. Step 3: **Yes**. (Set with $e^t \cos W_t$: $f - \frac{1}{2}f \neq 0$ → No.)

13 ☆ $E[h(W_t) \mid \mathcal{F}_s]$, $s < t$ (future given present).

FORMULA

$W_t = W_s + \Delta$, $\Delta \sim N(0, t-s)$ independent of $\mathcal{F}_s$: treat W_s as a constant, average over Δ.

h	$E[h(W_t) \mid \mathcal{F}_s]$
W_t	W_s
W_t^2	$W_s^2 + (t-s)$
$e^{\lambda W_t}$	$e^{\lambda W_s + \lambda^2(t-s)/2}$

USE WHEN

"Find $E[W_3^2 \mid \mathcal{F}_1]$", "filtration expectation".

HOW TO PLUG

1. Write $W_t = W_s + \Delta$.
2. Expand; W_s is a constant.
3. $E\Delta = 0$, $E\Delta^2 = t - s$, $Ee^{\lambda\Delta} = e^{\lambda^2(t-s)/2}$.

CAVEATS

Only for a LATER time given an earlier one. Given the future ($W_t = b$, want W_s, $s < t$): card 16.

EXAMPLE (SHEET)

$E[W_3^2 \mid \mathcal{F}_1] \to \Delta = W_3 - W_1 \sim N(0,2)$: $E[(W_1 + \Delta)^2 \mid \mathcal{F}_1] = W_1^2 + 0 + 2 = W_1^2 + 2$.

14 ⭐ Distribution of $\int_0^T f(t)\,dW_t$, and the Itô isometry.

FORMULA

$$f \text{ deterministic}: \quad \int_0^T f(t)\,dW_t \sim N\Big(0,\ \int_0^T f(t)^2\,dt\Big)$$

Isometry (any adapted H): $E\big[(\int_0^T H\,dW)^2\big] = \int_0^T E[H_t^2]\,dt$. Mean always 0. Shortcut: $\int_0^1 t^k dW_t \sim N(0, \frac{1}{2k+1})$.

USE WHEN

"Compute the distribution of $\int_0^1 t\,dW_t$", "$\mathrm{Var}(\int \ldots dW)$".

HOW TO PLUG

1. Check the integrand has no W in it.
2. Square it, integrate dt → the variance.
3. Answer $N(0, \text{that})$.

CAVEATS

Integrand containing W (e.g. $\int W\,dW$): mean 0 and isometry still hold, but it is NOT normal ($\int_0^t W\,dW = \frac{1}{2}W_t^2 - \frac{t}{2}$).

> EXAMPLE — CT-1'26 Q1 / CT-2'25 Q7
> $\int_0^1 t\,dW_t \to f = t$ deterministic; $\int_0^1 t^2 dt = \frac{1}{3} \to N(0, \frac{1}{3})$. CT-2'25: $\int_0^1 t^2 dW_t \to \int_0^1 t^4 dt = \frac{1}{5} \to N(0, \frac{1}{5})$.

15 Probabilities for W_t (and BM started at x_0); the heat equation.

FORMULA

$P(W_t \le x) = \Phi\big(\frac{x}{\sqrt{t}}\big)$; density $\frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}$. General $X \sim N(m, v)$: $P(X \le x) = \Phi\big(\frac{x-m}{\sqrt{v}}\big)$.
Transition density $p(t; x, y) = \frac{1}{\sqrt{2\pi t}} e^{-(y-x)^2/(2t)}$ solves $\partial_t p = \frac{1}{2}\partial_{yy} p$.

USE WHEN

"$P(W(4) > 3)$", "BM started at 5, $P(W(1) < 7)$", "show the density solves the heat equation".

HOW TO PLUG

1. Find the mean m (start point, drift) and variance v (t, or $\sigma^2 t$).
2. Standardize: $\frac{x-m}{\sqrt{v}}$.
3. Look up Φ.

CAVEATS

Divide by $\sqrt{t}$, NEVER by t. This is the most common lost mark.

> EXAMPLE (SHEET)
> $P(W(4) > 3) \to m = 0$, $\sqrt{v} = 2$: $1 - \Phi(\frac{3}{2}) = 0.0668$. BM from 5: $P(W(1) < 7) = \Phi(\frac{7-5}{1}) = \Phi(2) = 0.9772$.

16 ⭐ Conditioning on a value of BM: forward, backward (bridge), pinned.

FORMULA

(X, Y) jointly normal: $X \mid Y{=}y \sim N\Big(\mu_X + \frac{\mathrm{Cov}}{\mathrm{Var}\,Y}(y - \mu_Y),\ \ \mathrm{Var}\,X - \frac{\mathrm{Cov}^2}{\mathrm{Var}\,Y}\Big)$

Given	Law
forward: $W_t \mid W_s{=}a,\ s<t$	$N(a,\ t-s)$
backward: $W_s \mid W_t{=}b,\ s<t$	$N\left(\frac{s}{t}b,\ \frac{s(t-s)}{t}\right)$
pinned $W_0{=}a,\ W_t{=}b$	$N\left(a+\frac{s}{t}(b-a),\ \frac{s(t-s)}{t}\right)$
$Y=\sigma W,\ Y_t{=}b$	same mean, variance $\times\sigma^2$

☆ MOST GENERAL: $Y_T = X_0 + MT + \Sigma W_T$ TWO KNOWN VALUES, T IN BETWEEN

$s_1 < t < s_2,\ Y_{s_1} = a_1,\ Y_{s_2} = a_2$:

$$Y_t \mid (Y_{s_1}{=}a_1,\ Y_{s_2}{=}a_2) \ \sim\ N\Big(\underbrace{\tfrac{s_2-t}{s_2-s_1}}_{\text{weight on } a_1}\ a_1 + \underbrace{\tfrac{t-s_1}{s_2-s_1}}_{\text{weight on } a_2}\ a_2,\ \ \sigma^2\,\frac{(t-s_1)(s_2-t)}{s_2-s_1}\Big)$$

The drift μ and the start x_0 do NOT appear: once both ends are pinned, the drift is already inside a_1 and a_2. Plain W is $x_0=0,\ \mu=0,\ \sigma=1$.

What is known	Law of Y_t
two values, $s_1 < t < s_2$	formula above
only the start ($Y_0 = x_0$)	$N(x_0+\mu t,\ \sigma^2 t)$
latest known $Y_s = a,\ t > s$ (forward)	$N\big(a+\mu(t-s),\ \sigma^2(t-s)\big)$, here μ DOES matter
start x_0 and $Y_{s_2} = a_2,\ t < s_2$	formula above with $s_1 = 0,\ a_1 = x_0$

Intuition: a weighted average. The mean is an average of the two known values, and the two weights add to 1. Each value gets the weight of the distance to the OTHER end. So the closer t is to s_1, the more a_1 counts. At $t = s_1$ the weights are $(1, 0)$ and you get a_1; at $t = s_2$ you get a_2; at the midpoint it's $\frac{a_1+a_2}{2}$. On a graph, it's the straight line joining (s_1, a_1) and (s_2, a_2), read at t.

Variance = σ^2 × (distance to the left end) × (distance to the right end) ÷ (gap). It is 0 at both ends (you know the value there) and largest at the midpoint, where it is $\sigma^2\,\frac{s_2-s_1}{4}$.

Only the nearest known values matter (Markov): any value known before s_1 or after s_2 is irrelevant. Every row of the table above is this formula: backward is $s_1 = 0,\ a_1 = 0$; pinned is $s_1 = 0$.

- t must lie BETWEEN the two known times. If $t > s_2$, only the latest value counts: forward row $N\big(a_2+\mu(t-s_2),\ \sigma^2(t-s_2)\big)$.
- Between two known values, drop μ; going forward past the last known value, keep μ.
- Check whether they gave σ or σ^2.
- Not for GBM: there, take logs first ($\ln Y$ is BM with drift), apply this to $\ln Y$, then convert back.

EXAMPLE (PRACTICE)

$W_1 = 2,\ W_5 = 10$; law of W_2? → $s_1 = 1,\ s_2 = 5$, gap 4. Weights: on a_1: $\frac{5-2}{4} = \frac{3}{4}$; on a_2: $\frac{2-1}{4} = \frac{1}{4}$. Mean $\frac{3}{4}(2) + \frac{1}{4}(10) = 1.5 + 2.5 = 4$. Variance $\frac{(1)(3)}{4} = 0.75$. So $W_2 \sim N(4,\ 0.75)$.

Check with the straight line: slope $\frac{10-2}{4} = 2$ per unit time, so at $t = 2$: $2 + 2(1) = 4$ ✓.

With drift and volatility: $Y_t = 2t + 3W_t,\ Y_1 = 5,\ Y_5 = 13$, law of Y_2? → same weights $\frac{3}{4}, \frac{1}{4}$: mean $3.75 + 3.25 = 7$ (the drift 2 is not used); variance $9 \cdot \frac{1\cdot3}{4} = 6.75$.

Sheet: $Y = 3W,\ Y_{10} = 12$, law of Y_4 → $s_1 = 0,\ a_1 = 0$: mean $\frac{4}{10}(12) = 4.8$, variance $9 \cdot \frac{4\cdot6}{10} = 21.6$.

Bridge $B(s) = W(s) - \frac{s}{t}W(t)$: $\mathrm{Cov}(B_u, B_v) = \min(u, v) - \frac{uv}{t}$.

USE WHEN

"Given $W(3) = 2$, find $E[W(1)]$ / $P(W(1) < x)$", "Brownian bridge".

HOW TO PLUG

1. Which time is known: earlier (forward) or later (backward)?
2. Pick the row; plug s, t, the known value.
3. For a probability, standardize with card 15.

CAVEATS

Forward: mean stays, variance grows. Backward: mean is the straight line, variance is 0 at both ends. Don't swap them.

EXAMPLE (SHEET)

"Given $W(3) = 2$, find $E[W(1)]$" → backward, $s = 1$, $t = 3$, $b = 2$: mean $\frac{1}{3}(2) = \frac{2}{3}$, variance $\frac{1 \cdot 2}{3} = \frac{2}{3}$. "Given $W(1) = 2$, $E[W(3)]$" → forward → 2.

17 ☆ **BM with drift $X_t = \mu t + \sigma W_t$: which barrier first, max ever, exit time, best exercise level.**

FORMULA

$X_t \sim N(\mu t, \sigma^2 t)$. Let $\theta = 2\mu/\sigma^2$. Up barrier $A > 0$, down barrier $-B < 0$, start 0.

Question	Formula
$P(\text{hit } A \text{ before } -B)$, $\mu \neq 0$	$\dfrac{1 - e^{-\theta B}}{1 - e^{-\theta(A+B)}}$
same, $\mu = 0$	$\frac{B}{A+B}$; and $P(-B \text{ first}) = \frac{A}{A+B}$
$P(\max_{t \geq 0} X_t \geq a)$, $\mu < 0$	$e^{-2\lvert\mu\rvert a/\sigma^2}$ ($= 1$ if $\mu \geq 0$)
$E[\text{exit time}]$, $\mu = 0$	AB/σ^2
option to buy at K, drift $-\mu$: when to exercise	first time price hits $x^* = K + \frac{\sigma^2}{2\mu}$
$T = \min\{t : W_t = a + bt\}$, $ab < 0$	$E[T] = -a/b$ (OST: $EW_T = 0$)

USE WHEN

"Reaches -2 before 1", "drift coefficient -1… when to exercise", "$W_t = 2 - 3t$".

HOW TO PLUG

1. Read μ and σ^2 (careful which one they gave).
2. Shift so the start is 0: start x with barriers $a < x < b \to A = b - x$, $B = x - a$.
3. $\mu = 0$ or not? Pick the row.

CAVEATS

- $\mu = 0$ makes the exponential formula $0/0$: use $\frac{B}{A+B}$.
- "Volatility $\sigma = 0.3$" is σ, not σ^2.
- Sloping line needs $ab < 0$, else $E[T] = \infty$.

EXAMPLE — MIDSEM'24 Q1(C)

$\mu = 0$, $\sigma^2 = 1$: $P(\text{reaches } -2 \text{ before } 1) \to A = 1$, $B = 2$, $\mu = 0$ row: $\frac{A}{A+B} = \frac{1}{3}$.
Midsem'25 Q1(a): buy at $K = 100$, drift -1 ($\mu = 1, \sigma = 1$) → $x^* = 100 + \frac{1}{2} = 100.5$. Endsem'25 Q1(d): $W_t = 2 - 3t \to 0 = 2 - 3E[T] \to \frac{2}{3}$.

18 ⭐ GBM master: mean, moments, conditional mean, MG condition, median, $P(Y_t > K)$.

FORMULA

$\ln Y_t = \ln Y_0 + mt + \sigma W_t \sim N(\ln Y_0 + mt,\ \sigma^2 t)$, where

How it is given	m	$E[Y_t]$
(R) $Y_t = Y_0 e^{\mu t + \sigma W_t}$, "log of price is BM with drift μ"	μ	$Y_0 e^{(\mu+\sigma^2/2)t}$
(S) $dY = \mu Y\,dt + \sigma Y\,dW$, "expected return μ"	$\mu - \frac{\sigma^2}{2}$	$Y_0 e^{\mu t}$

$E[Y_t^n] = Y_0^n e^{(nm+n^2\sigma^2/2)t}$; $\mathrm{Var} = E[Y_t]^2(e^{\sigma^2 t}-1)$; $E[Y_t \mid \mathcal{F}_s]$: replace $Y_0 \to Y_s$, $t \to t-s$;
MG $\iff m + \frac{\sigma^2}{2} = 0$; median $Y_0 e^{mt}$; $P(Y_t > K) = 1 - \Phi\left(\frac{\ln(K/Y_0) - mt}{\sigma\sqrt{t}}\right)$.

USE WHEN

"GBM with drift μ and volatility σ, $Y(0) = 5$, find $E[Y(10)]$", "is the price a martingale", "probability price exceeds K".

HOW TO PLUG

1. Decide (R) or (S) from the wording; if unclear, do both.
2. Get m.
3. Plug into the line you need.

CAVEATS

- "Drift parameter μ" is ambiguous: write BOTH answers, labelled. The CT-1 key accepted either.
- $P(Y_t > K)$: take logs first, then standardize.

EXAMPLE — CT-1'26 Q6

$\mu = 0.01, \sigma = 0.3, Y(0) = 5, E[Y(10)] \to$ (R): $5e^{(0.01+0.045)10} = 5e^{0.55}$; (S): $5e^{0.01\cdot 10} = 5e^{0.1}$.
Midsem'25 Q1(b): $\mu = 0.01, \sigma = 0.2, Y_0 = 10, t = 100 \to$ (R) $10e^{(0.01+0.02)100} = 10e^3$; (S) $10e^1$.

19 ⭐ Running max M_t and hitting time τ_a.

FORMULA

$\{\tau_a \le t\} = \{M_t \ge a\}$ $(a > 0)$, and $M_t \overset{d}{=} |W_t|$.

Quantity	Value
$P(M_t \ge a) = P(\tau_a \le t)$ (cdf of τ_a)	$2\left(1 - \Phi(\frac{a}{\sqrt{t}})\right)$
pdf of τ_a	$\frac{\lvert a\rvert}{\sqrt{2\pi}\,t^{3/2}} e^{-a^2/(2t)}$
$P(\tau_a < \infty)$, $E[\tau_a]$	$1,\ \infty$
$E[M_t]$, $E[M_t^2]$	$\sqrt{2t/\pi},\ t$
$E[e^{-\lambda\tau_a}]$	$e^{-a\sqrt{2\lambda}}$
$P(\min_{u\le t} W_u \le -a)$	same as $P(M_t \ge a)$

USE WHEN

"Find the cdf and pdf of τ_a", "$P(T_a < \infty)$", "expected highest level reached".

HOW TO PLUG

1. Max question ↔ hitting-time question: rewrite to whichever row you know.
2. Plug a and $\sqrt{t}$.
3. pdf: differentiate the cdf.

CAVEATS

- For $a < 0$ use $|a|$ (symmetry).
- These are for M alone. If W_t also appears: card 20.

EXAMPLE — MIDSEM'25 Q5(A), MIDSEM'24 Q1(F)

cdf of $\tau_a \to 2(1-\Phi(a/\sqrt{t}))$; pdf by $\frac{d}{dt}$; at $a=2, t=4$: $\frac{2}{\sqrt{2\pi}\cdot 8}e^{-1/2} = 0.0605$. "$P(T_a < \infty)$" $\to 1$.

20 ☆ **Joint (W_t, M_t): $P(a < W_t \le b,\ c < M_t \le d)$.**

FORMULA

$$G(b,c) := P(W_t \le b,\ M_t \ge c) = \begin{cases} \Phi(\frac{b}{\sqrt{t}}) & c \le 0 \\ 1 - \Phi(\frac{2c-b}{\sqrt{t}}) & c > 0,\ b \le c \\ 1 + \Phi(\frac{b}{\sqrt{t}}) - 2\Phi(\frac{c}{\sqrt{t}}) & 0 < c < b \end{cases}$$

$$P = [G(b,c) - G(a,c)] - [G(b,d) - G(a,d)]$$

USE WHEN

"$P(1 < W(4) < 3,\ 2 < M(4) < 3)$", barrier-option probabilities.

HOW TO PLUG

1. First: $a \ge d$ or $d \le 0$ → answer 0.
2. Compute the four G's, choosing the branch by comparing b with c each time.
3. Combine.

CAVEATS

The $b > c$ case is NOT zero and NOT the reflection formula: use branch 3.

EXAMPLE — MIDSEM'25 Q5(B)

$P(1 < W_4 < 3,\ 2 < M_4 < 3)$, $\sqrt{t} = 2, a = 1, b = 3, c = 2, d = 3$.
$G(3,2)$: $b > c \to 1 + \Phi(1.5) - 2\Phi(1) = .2505$. $G(1,2) = 1 - \Phi(\frac{4-1}{2}) = .0668$. $G(3,3) = 1 - \Phi(1.5) = .0668$.
$G(1,3) = 1 - \Phi(2.5) = .0062$.
$[.2505 - .0668] - [.0668 - .0062] = 0.1231$.

3. Itô's formula

21 ☆ **Itô's formula (general) and the multiplication table.**

FORMULA

$dt \cdot dt = 0, \quad dt \cdot dW = 0, \quad dW \cdot dW = dt$. For $dX = a\,dt + b\,dW$: $(dX)^2 = b^2\,dt$.

$$df(t, X_t) = \partial_t f\,dt + \partial_x f\,dX_t + \tfrac{1}{2}\partial_{xx} f\,b^2\,dt$$

For $X = W$: $df = \left(\partial_t f + \tfrac{1}{2}\partial_{xx} f\right)dt + \partial_x f\,dW$.
Product: $d(XY) = X\,dY + Y\,dX + b_X b_Y\,dt$.

USE WHEN

"Compute dY_t", "derive Itô's formula", every SDE question.

HOW TO PLUG

1. Identify a and b of dX (for W: $a = 0$, $b = 1$).
2. Compute $\partial_t f$, $\partial_x f$, $\partial_{xx} f$.
3. Plug; collect the dt part (drift) and the dW part (diffusion).

CAVEATS

- No ∂_{tt} term. Only the x-variable gets the $\frac{1}{2}$ correction.
- If f is linear in x (e.g. $g(t)W_t$), the correction is 0.

EXAMPLE — MIDSEM'25 Q2(B)

"Derive Itô's formula" [2 marks] → Taylor: $df = f_t\,dt + f_x\,dX + \frac{1}{2}f_{xx}(dX)^2 + \ldots$; table: $(dW)^2 = dt$ because $\Delta W \sim \sqrt{\Delta t}$; dt^2, $dt\,dW$ and higher orders vanish → the formula.

22 ☆ **The differential dictionary.**

FORMULA

Y_t	dY_t
$g(W)$ (any)	$g'(W)\,dW + \frac{1}{2}g''(W)\,dt$
W^n	$nW^{n-1}dW + \frac{1}{2}n(n-1)W^{n-2}dt;\ d(W^2) = 2W\,dW + dt$
$\sin W$	$\cos W\,dW - \frac{1}{2}\sin W\,dt$
$\cos W$	$-\sin W\,dW - \frac{1}{2}\cos W\,dt$
$e^{\sigma W}$	$\sigma e^{\sigma W}dW + \frac{\sigma^2}{2}e^{\sigma W}dt$
$e^{\sigma W - \sigma^2 t/2}$	$\sigma Y\,dW$ (zero drift)
$e^{\lambda t}g(W)$	$e^{\lambda t}\left(\lambda g + \frac{1}{2}g''\right)dt + e^{\lambda t}g'(W)\,dW$
$g(t)W$ (tW, $e^t W$)	$g'(t)W\,dt + g(t)\,dW$
$e^{\mu t + \sigma W}$	$(\mu + \frac{\sigma^2}{2})Y\,dt + \sigma Y\,dW$

USE WHEN

1-mark "compute $d\sin(W_t)$", "$d(e^t W_t)$".

HOW TO PLUG

1. Match the row.
2. For $g(W)$: diffusion $= g'$, drift $= \frac{1}{2}g''$. Two derivatives, done.

CAVEATS

Ordinary calculus forgets the $\frac{1}{2}g''\,dt$. Never drop it (except when $g'' = 0$).

EXAMPLE — CT-2'25 Q8, ENDSEM'25 Q1(B)

$d\sin W_t \to g' = \cos$, $\frac{1}{2}g'' = -\frac{1}{2}\sin \to \cos W_t\, dW_t - \frac{1}{2}\sin W_t\, dt$.

$d(e^t W_t) \to$ row $g(t)W$, $g = e^t \to e^t W_t\, dt + e^t\, dW_t$.

23 ☆ Evaluate any Itô integral $\int_0^t g(s, W_s)\, dW_s$ using Itô's formula.

FORMULA

Pick $F(t,x)$ with $\partial_x F = g$. Itô (card 21, $X = W$): $dF = \partial_t F\, dt + \partial_x F\, dW + \frac{1}{2}\partial_{xx} F\, dt$. Integrate from 0 to t and move things across:

$$\int_0^t g(s, W_s)\, dW_s = F(t, W_t) - F(0,0) - \int_0^t \left(\partial_s F + \tfrac{1}{2}\partial_{xx} F\right)(s, W_s)\, ds$$

USE WHEN

"Evaluate / compute $\int_0^t W_s\, dW_s$", "$\int_0^t s\, dW_s$", "$\int_0^t W_s^2\, dW_s$", "$\int e^{W_s} dW_s$", "show $\int \ldots dW = \ldots$".

HOW TO PLUG

1. Name the integrand $g(s,x)$: replace W_s by x.
2. Antiderivative in x only (treat s as a constant): $F = \int g\, dx$.
3. Compute $\partial_s F$ and $\frac{1}{2}\partial_{xx} F$.
4. Write: answer $= F(t, W_t) - F(0,0) - \int_0^t (\partial_s F + \frac{1}{2}\partial_{xx} F)\, ds$.

Integral	$F(s,x)$	$\partial_s F + \frac{1}{2}\partial_{xx} F$	Answer
$\int_0^t W_s\, dW_s$	$\frac{x^2}{2}$	$0 + \frac{1}{2}$	$\frac{1}{2}W_t^2 - \frac{1}{2}t$
$\int_0^t s\, dW_s$	sx	$x + 0$	$tW_t - \int_0^t W_s\, ds$
$\int_0^t W_s^2\, dW_s$	$\frac{x^3}{3}$	$0 + x$	$\frac{1}{3}W_t^3 - \int_0^t W_s\, ds$
$\int_0^t e^{W_s} dW_s$	e^x	$\frac{1}{2}e^x$	$e^{W_t} - 1 - \frac{1}{2}\int_0^t e^{W_s} ds$
$\int_0^t \cos W_s\, dW_s$	$\sin x$	$-\frac{1}{2}\sin x$	$\sin W_t + \frac{1}{2}\int_0^t \sin W_s\, ds$
$\int_0^t e^{W_s - s/2} dW_s$	$e^{x - s/2}$	$-\frac{1}{2}F + \frac{1}{2}F = 0$	$e^{W_t - t/2} - 1$
$\int_0^t f(s)\, dW_s$, f deterministic	$f(s)x$	$f'(s)x$	$f(t)W_t - \int_0^t f'(s)W_s\, ds$ (integration by parts)

CAVEATS

- Itô's formula is $df = \partial_t f\, dt + \partial_x f\, dX + \frac{1}{2}\partial_{xx} f\, (dX)^2$: the $\frac{1}{2}$ is only on the second-derivative term, never on $\partial_x f\, dX$.
- The leftover $\int_0^t (\ldots)\, ds$ is an ordinary ds-integral of the path. If it doesn't vanish, leave it in the answer; that is a complete answer.
- $F(0,0)$ uses $W_0 = 0$. Don't drop the constant (e.g. $e^0 = 1$).
- For a variance, don't evaluate: use the isometry (card 14). The integral is the LEFT-endpoint limit $\lim \sum g(t_{i-1})(W_{t_i} - W_{t_{i-1}})$; a midpoint rule gives a different (Stratonovich) integral.

EXAMPLE — THE CLASSIC (ENDSEM/MIDSEM SHORT QUESTION)

Evaluate $\int_0^t W_s\, dW_s$.

Step 1: $g = x$. Step 2: $F = \frac{x^2}{2}$. Step 3: $\partial_s F = 0$, $\frac{1}{2}\partial_{xx} F = \frac{1}{2}$. Step 4: $\frac{W_t^2}{2} - 0 - \int_0^t \frac{1}{2}\, ds = \frac{1}{2}W_t^2 - \frac{1}{2}t$.

Check: mean $\frac{t}{2} - \frac{t}{2} = 0$ ✓ (every Itô integral has mean 0).

Endsem'25 Q1(a): $\mathrm{Var}\left(\int_0^1 t^4 dW_t\right)$ → isometry, not this card: $\int_0^1 t^8 dt = \frac{1}{9}$.

4. SDEs and interest rate models

24 ☆ **Mean and variance of a solution of $dX = a(t,X)\,dt + b(t,X)\,dW$ without solving it.**

FORMULA

$$\frac{d}{dt}E[X_t] = E[a(t,X_t)], \qquad \frac{d}{dt}E[X_t^2] = E\left[2X_t a + b^2\right]$$

then $\mathrm{Var} = E[X^2] - (EX)^2$.

USE WHEN

"Find $E(X_t)$ and $\mathrm{Var}(X_t)$" for a given SDE.

HOW TO PLUG

1. Read a, b, x_0.
2. Mean ODE: $m' = E[a]$, $m(0) = x_0$; solve.
3. Second moment ODE: plug a, b, replace $E[X]$ by $m(t)$; integrate.
4. Subtract m^2.

CAVEATS

Closes only if a is linear in X and b^2 linear/constant in X. If $a = X^2$, $E[X^2] \neq m^2$ and the ODE does not close.

EXAMPLE — MIDSEM'25 Q4(A)

$dX = dt + 2\sqrt{X}\,dW$, $X_0 = 0$. → $a = 1$, $b^2 = 4X$.

Mean: $m' = 1 \to EX_t = t$. Second moment: $\frac{d}{dt}EX^2 = E[2X(1) + 4X] = 6t \to EX^2 = 3t^2$. Variance: $3t^2 - t^2 = 2t^2$.

25 ☆ **Is an Itô process a martingale? And $E[f(W_t)]$ through a martingale.**

FORMULA

$dX = a\,dt + b\,dW$ is a MG $\iff$ $a \equiv 0$.

If $\lambda f + \frac{1}{2}f'' = 0$, then $e^{\lambda t} f(W_t)$ is a MG, so $E[f(W_t)] = f(0)\,e^{-\lambda t}$.

USE WHEN

"Is Y_t a martingale?"; "using part (i), evaluate $E[\cos W_t]$".

HOW TO PLUG

1. Apply Itô (card 21) to get dY.
2. Read the dt coefficient: $0 \to$ MG.
3. For $E[f(W_t)]$: MG ⇒ $E[Y_t] = Y_0$, then divide by $e^{\lambda t}$.

CAVEATS

Zero drift needs the right λ exactly (for $\sin, \cos$: $\frac{1}{2}$).

EXAMPLE — MIDSEM'25 Q4(B)

$X_t = e^{t/2}\cos W_t$. (i) dX: $f_t = \frac{1}{2}f$, $f_x = -e^{t/2}\sin x$, $f_{xx} = -f \to dX = -e^{t/2}\sin W_t\, dW_t$ (drift 0). (ii) MG → $e^{t/2}E[\cos W_t] = \cos 0 = 1 \to E[\cos W_t] = e^{-t/2}$.

Endsem'25 Q1(c): is $e^t W_t$ a MG? drift $e^t W_t \neq 0 \to$ **No**.

26 "Show $X_t = \ldots$ solves the SDE", and when is the solution Gaussian?

FORMULA

Verify: Itô on the candidate, rewrite in terms of X_t, compare with the SDE, check X_0.

Gaussian $\iff$ the dW coefficient contains no X.

USE WHEN

"Show that $X_t = W_t^2$ is a solution", "is X_t normally distributed?".

HOW TO PLUG

1. Compute dX_t of the candidate.
2. Replace every W-expression by X.
3. Match dt and dW parts; check X_0.

CAVEATS

Gaussian: BM with drift, OU, Vasicek. Not Gaussian: GBM (log-normal), CIR, $dX = dt + 2\sqrt{X}\, dW$.

EXAMPLE — MIDSEM'25 Q4(A)

Show $X_t = W_t^2$ solves $dX = dt + 2\sqrt{X}\, dW$, $X_0 = 0 \to$ Itô: $d(W_t^2) = dt + 2W_t\, dW_t$. With $X_t = W_t^2$, $\sqrt{X_t} = |W_t|$, so $2W_t\, dW_t$ matches $2\sqrt{X_t}\, dW_t$ ✓ (the paper's accepted answer); $X_0 = W_0^2 = 0$ ✓.

27 ⭐ Linear SDE master $dX = \kappa(b - X)\, dt + \sigma\, dW$: OU, Vasicek, CIR.

FORMULA

Multiply by $e^{\kappa t}$ (integrating factor):

$$X_t = b + (x_0 - b)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s$$

$$X_t \sim N\Big(b + (x_0 - b)e^{-\kappa t},\ \tfrac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})\Big) \xrightarrow{t\to\infty} N\big(b,\ \tfrac{\sigma^2}{2\kappa}\big)$$

Model	Read off
OU $dX = -rX\, dt + \sigma dW$	$b = 0$, $\kappa = r$
Vasicek $dr = \kappa(b - r)dt + \sigma dW$	as is; rates can go negative
CIR $dr = \kappa(b - r)dt + \sigma\sqrt{r}\, dW$	same MEAN only; not normal; $r \geq 0$; Feller $2\kappa b \geq \sigma^2 \Rightarrow r > 0$
$dX = \mu dt + \sigma dW$	$X_t = x_0 + \mu t + \sigma W_t \sim N(x_0 + \mu t, \sigma^2 t)$

USE WHEN

"Express X_t as a stochastic integral", "is it OU?", "mean and variance", "Vasicek/CIR".

HOW TO PLUG

1. Rewrite the drift as $\kappa(b - X)$; read κ, b, σ, x_0.
2. Plug into the solution / mean / variance.

CAVEATS

Works only if the noise has no X. For CIR only the mean formula holds (use card 24).

EXAMPLE — MIDSEM'24 Q4(A)

$dX = -X\,dt + dW$, $X_0 = x_0 \to \kappa = 1$, $b = 0$, $\sigma = 1$.

$X_t = x_0 e^{-t} + \int_0^t e^{-(t-s)} dW_s$; OU? **Yes** (mean-reverting to 0); $EX_t = x_0 e^{-t}$; $\mathrm{Var} = \frac{1}{2}(1 - e^{-2t})$.

28 GBM SDE $dN = rN\,dt + \alpha N\,dW$: solution, mean, median.

FORMULA

Itô on $\ln N$: $d \ln N = (r - \frac{\alpha^2}{2})dt + \alpha\,dW$, so

$$N_t = N_0 \exp\left\{(r - \tfrac{\alpha^2}{2})t + \alpha W_t\right\}, \qquad EN_t = N_0 e^{rt}, \quad \text{median } N_0 e^{(r-\alpha^2/2)t}$$

USE WHEN

"Solve $dN = rN\,dt + \alpha N\,dW$", "is $e^{W_t - t/2}$ a martingale".

HOW TO PLUG

1. Read r and α.
2. Write the exponent $(r - \frac{\alpha^2}{2})t + \alpha W_t$.
3. MG $\iff$ $r = 0$ (equivalently exponent $= \alpha W_t - \frac{\alpha^2}{2} t$).

CAVEATS

Taking $\ln$ needs $N_0 > 0$. Log-normal, not normal.

EXAMPLE — MIDSEM'24 Q4(B)

Is $X_t = e^{W_t - t/2}$ a MG? → it is GBM with $r = 0$, $\alpha = 1$: $dX = X_t\,dW_t$, zero drift → **Yes**.

5. No-arbitrage, forwards, calls and puts

29 Arbitrage: definition, NA condition, the arbitrage theorem.

FORMULA

Arbitrage: $V_0 = 0$ (or ≤ 0), $V_T \ge 0$ in EVERY state, $V_T > 0$ in some state.

Theorem: no arbitrage $\iff$ a risk-neutral (EMM) probability exists. Binomial: NA $\iff$ $d < 1 + r < u$.

USE WHEN

"Show the market is arbitrage-free", "define arbitrage", "find the arbitrage".

HOW TO PLUG

1. Compute $u = S_u/S_0$, $d = S_d/S_0$, $1 + r$.
2. Check $d < 1 + r < u \rightarrow$ arbitrage-free.
3. If $1 + r \geq u$: short stock, lend. If $1 + r \leq d$: borrow, buy stock.

CAVEATS

"Every state" means with probability 1, not on average. No real probability enters.

EXAMPLE — ENDSEM'25 Q5(B)(I)

$S_0 = 5$, $S_1 \in \{6, \frac{40}{9}\}$, $r = \frac{1}{9} \rightarrow u = \frac{6}{5}$, $d = \frac{8}{9}$, $1 + r = \frac{10}{9}$; $\frac{8}{9} < \frac{10}{9} < \frac{6}{5}$ ✓ → **arbitrage-free**.

30 ☆ **Payoffs, forward price, put–call parity, American vs European.**

FORMULA

Contract	Payoff at T
long forward at K	$S_T - K$ (obligation, can be negative)
call	$(S_T - K)^+$
put	$(K - S_T)^+$

Fair forward price $K = S_0(1 + r)^T$ (or $S_0 e^{rT}$). Value today of a forward at K: $S_0 - K(1 + r)^{-T}$.

$$C - P = S_0 - K(1 + r)^{-T} \quad (\text{continuous: } S_0 - Ke^{-rT})$$

USE WHEN

"Find K so the forward costs nothing", "price the put" after the call, "short note American vs European".

HOW TO PLUG

1. Forward: $K = S_0(1 + r)^T$.
2. Have C, need P: $P = C - S_0 + K(1 + r)^{-T}$.

CAVEATS

- Parity: European only, same K, same T.
- American call = European call only with no dividends ($r \geq 0$). American put can be strictly dearer.

EXAMPLE — MIDSEM'25 Q2(A)

"Short note on American and European calls" → European: exercise only at T. American: at any stopping time $\leq T$, so American $\geq$ European. With no dividends early exercise of a call is never optimal, so they are equal; for a put it can be optimal. Forward: $S_0 = 100$, $r = 5\%$, $T = 1 \rightarrow K = 105$.

31 ☆ **The "no sure win" option price c^* (prices given as present values).**

FORMULA

$$p = \frac{S_0 - S_d}{S_u - S_d}, \qquad c^* = p\,(S_u - K)^+ + (1 - p)(S_d - K)^+$$

No sure win $\iff c = c^*$; arbitrage $\iff c \neq c^*$.

USE WHEN

"Price 50 goes to 25 or 100; option to buy at K costs c per share; for which c is there no sure win?" (3 papers in a row).

HOW TO PLUG

1. p from the stock.
2. Option payoffs in both states.
3. c^* = weighted average.

CAVEATS

- Prices already present values → no discounting. If an r is given, use card 32 instead.
- Never use the real-world probability.

EXAMPLE — MIDSEM'25 Q1(C)

$50 \to \{25, 100\}$, $K = 85$ → Step 1: $p = \frac{50-25}{100-25} = \frac{1}{3}$. Step 2: payoffs 15 and 0. Step 3: $c^* = \frac{1}{3}(15) = 5$.
CT-2'25 Q5: $40 \to \{25, 100\}$, $K = 75 \to p = \frac{1}{5}$, payoff $25 \to c = 5$.

6. Binomial and finite markets

32 ☆ **One-period binomial: NA, p^*, price, replicating portfolio.**

FORMULA

$$d < 1 + r < u, \qquad p^* = \frac{1 + r - d}{u - d}, \qquad V_0 = \frac{p^* V_u + (1 - p^*) V_d}{1 + r}$$

$$\Delta = \frac{V_u - V_d}{(u - d) S_0}, \qquad B = \frac{u V_d - d V_u}{(1 + r)(u - d)}, \qquad V_0 = \Delta S_0 + B$$

USE WHEN

"Price the call/put", "construct the replicating portfolio (Δ, B)".

HOW TO PLUG

1. $u = S_u / S_0$, $d = S_d / S_0$; check NA.
2. p^*.
3. Payoffs V_u, V_d.
4. V_0 by the formula; Δ, B; check $\Delta S_0 + B = V_0$.

CAVEATS

- Never use the real p for prices.
- $\Delta < 0$ for a put; $B < 0$ means borrowing.
- $E[S_t] = S_0(pu + (1-p)d)^t$ uses the REAL p (that's a different question).

EXAMPLE — CT-2'25 Q1

Put, $K = 225$, $S_0 = 200$, $u = 1.25$, $d = 0.25$, $r = 0.15$.
$p^* = \frac{1.15 - 0.25}{1.25 - 0.25} = 0.9$. $V_u = (225 - 250)^+ = 0$, $V_d = (225 - 50)^+ = 175$.
$V_0 = \frac{0.9(0) + 0.1(175)}{1.15} = 15.22$. $\Delta = \frac{0 - 175}{1(200)} = -0.875$, $B = \frac{1.25(175) - 0}{1.15} = 190.22$. Check: $-175 + 190.22 = 15.22$ ✓.

33 ☆ Multi-period binomial: backward induction, hedge at each node, self-financing.

FORMULA

At node $S_{t-1} = s$:

$$V_{t-1}[s] = \frac{p^* V_t[us] + (1-p^*)V_t[ds]}{1+r}, \qquad V_T = g(S_T)$$

$\Delta_t(s) = \dfrac{V_t[us] - V_t[ds]}{us - ds}$, $b_t(s) = \dfrac{uV_t[ds] - dV_t[us]}{(1+r)^t(u-d)}$ (number of bonds, bond price $(1+r)^t$).
Self-financing: $\Delta_t S_t + b_t(1+r)^t = \Delta_{t+1}S_t + b_{t+1}(1+r)^t$.
Direct: $V_0 = \frac{1}{(1+r)^T}\sum_{j=0}^{T}\binom{T}{j}(p^*)^j(1-p^*)^{T-j}g(u^j d^{T-j}S_0)$.

USE WHEN

"$T = 2$ or 3 periods, price the option and give the strategy at each node", "verify self-financing".

HOW TO PLUG

1. Draw the tree; leaves $S_0 u^j d^{T-j}$.
2. Payoffs at the leaves.
3. Step back one period at a time with the V_{t-1} formula.
4. At each node: Δ_t, b_t.

CAVEATS

- b_t counts bonds, its cash value is $b_t(1+r)^t$.
- Path-dependent payoffs: don't recombine the tree.

EXAMPLE (SHEET)

$S_0 = 100$, $u = 2$, $d = \frac{1}{2}$, $r = \frac{1}{4}$, call $K = 100$, $T = 2$.
$p^* = \frac{1.25-0.5}{1.5} = \frac{1}{2}$. Leaves $400, 100, 25$ → payoffs $300, 0, 0$. $V_1[200] = \frac{\frac{1}{2}(300)}{1.25} = 120$, $V_1[50] = 0$. $V_0 = \frac{\frac{1}{2}(120)}{1.25} = 48$. Direct: $\frac{\frac{1}{4}(300)}{1.5625} = 48$ ✓.

34 ☆ Finite single-period market: EMM, Theorems 2, 3, 4.

FORMULA

EMM: all $p_k^* > 0$, $\sum p_k^* = 1$, and for each stock i: $\sum_k p_k^* S_1^i(\omega_k) = (1+r)S_0^i$.

Thm	Statement
2	NA $\iff$ an EMM exists $\iff$ $S_0 \in D_0$ (strictly inside the hull of discounted payoffs)
3	$E^*[\frac{V_1}{1+r}] = V_0$ for any strategy; attainable X: price $X^* = E^*[\frac{X}{1+r}]$, same for every EMM
4	complete $\iff$ EMM unique (needs states N = assets $n+1$, $\det A \neq 0$)

USE WHEN

"Find all EMMs", "is the market arbitrage-free / complete?", "price the claim", 3 or more branches from one node.

HOW TO PLUG

1. Unknowns $p_1^* .. p_N^*$: one equation per stock + sum $= 1$.
2. Solve; keep only all-positive solutions. None → arbitrage. One → complete. A line → incomplete.
3. Price X: compute $E^*[\frac{X}{1+r}]$ with the free parameter left in. It cancels → attainable, one price; it doesn't → a price interval.

CAVEATS

Incomplete market: no single price for a non-attainable claim, only an interval.

EXAMPLE (SHEET)

$100 \to \{300, 200, 50\}$, $K = 150$, $r = 0 \to$ stock: $300p_1 + 200p_2 + 50p_3 = 100$, sum $= 1 \to p_2 = \frac{1-5p_1}{3}$, $p_1 \in (0, \frac{1}{5})$ (a line: incomplete). Call: $c = 150p_1 + 50p_2 = \frac{200p_1+50}{3} \to c \in (\frac{50}{3}, 30)$.

7. Martingales and claim pricing

35 ⭐ **The three facts of claim pricing, and Black–Scholes.**

FORMULA

(1) EMM with discounted price a MG $\Rightarrow$ NA. (2) Attainable X: $P_t(X) = E^*\left[(1+r)^{t-T}X \mid \mathcal{F}_t\right]$ (continuous: $e^{-r(T-t)}$). (3) EMM unique $\Rightarrow$ complete.

Black–Scholes: under P^* the drift $\mu \to r$, $S_t = S_0 e^{(r-\sigma^2/2)t+\sigma W_t^*}$.

$$C = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \quad d_1 = \frac{\ln(S_0/K) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

USE WHEN

"Short note on martingales and claim pricing", "is the BS market arbitrage-free / complete?", "price a call with BS".

HOW TO PLUG

1. Find the EMM (p^*, or $\mu \to r$).
2. Price = discounted E^* of the payoff.
3. BS: d_1, d_2, two Φ look-ups; put by parity.

CAVEATS

- The real drift μ never appears in a price.
- Binomial discounts with $(1+r)^{-T}$, BS with e^{-rT}. Don't mix.

EXAMPLE (SHEET)

$S_0 = K = 100$, $r = 0.05$, $\sigma = 0.2$, $T = 1 \to d_1 = \frac{0+0.07}{0.2} = 0.35$, $d_2 = 0.15$; $C = 100(0.6368) - 95.12(0.5596) = 10.45$; put $= 10.45 - 100 + 95.12 = 5.57$.